PROJECT DOCUMENT

Mitsue Project

Revenue & Payback Summary

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Mitsue Project — Revenue & Payback Summary

One-Page Funder-Facing Overview

All figures are illustrative early-stage estimates. Phase 1 feasibility studies (due M9, Dec 2026) will replace these with vetted figures anchored to real survey data.

Full detail in `mitsue_implementation_plan.md` — "Return on Investment" section.

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1. Revenue Streams

Biomass CHP (fuelled by sugi forest thinnings) is the **primary energy source**, providing 24/7 baseload electricity and heat. Solar, battery storage, and EV charging are complementary. The circular local energy economy — where the forest powers the village — underpins all revenue streams.

Revenue / Savings Category	Year 1	Year 5	Year 10	Notes
Data center hosting fees	¥0	¥15–30M	¥40–80M	Edge computing / AI specialist workloads
Electricity sales (FIT/FIP)	¥0	¥5–15M	¥10–30M	Surplus from biomass CHP + solar generation
EV charging fees	¥0	¥1–3M	¥3–8M	Growing as EV fleet expands
Carbon credits (J-Credit)	¥0	¥1–3M	¥3–10M	From native forest restoration
Forestry products (timber, lumber)	¥0	¥3–8M	¥10–20M	Beyond fuel-residue use
Education / consulting ("playbook")	¥0	¥1–3M	¥3–8M	Replication licensing and training
Total annual revenue (illustrative)	**¥0**	**¥28–67M**	**¥74–166M**	

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2. Cost Displacement

The project *replaces or avoids* significant existing costs:

- **Energy imports**: Mitsue residents and businesses currently import ~100% of electricity. Local generation displaces approximately **¥40–60M/yr** of energy spending leaving the village economy.
- **Former Sugano school maintenance burden**: The Koryukan costs the village an estimated **¥3–8M/yr** in basic upkeep with no return. Active reuse as a data center hub converts a stranded asset into a productive facility.
- **Forest management liability**: Untended sugi plantations impose ecological costs (pollen, biodiversity loss) and physical risks (landslide, fire). Active management converts this liability into feedstock and timber

revenue.

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3. Payback Framework

Metric	Value
Total capital deployed (Phases 1–3) by Year 5	¥200–290M
Annual revenue by Year 5	¥28–67M
Operating costs by Year 5 (est. 60% of revenue)	¥18–35M
Net annual surplus by Year 5	**¥10–32M**
Approximate capital payback period	**10–18 years** (depending on grant vs. loan vs. revenue-financing mix)
Break-even / net-surplus year (base case)	~Year 5

Once net surplus is achieved, reinvestment priority: (1) forestry scale-up → (2) data-center expansion → (3) replication support for other villages.

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4. Honest Caveat

These are **framework figures, not a forecast**. They assume successful execution of all three project elements (biomass CHP energy, data center, forestry) at target scale. Phase 1 feasibility studies will sharpen these numbers significantly. Real-world performance — particularly biomass CHP sizing, data center occupancy, and FIT/FIP tariff outcomes — will determine actual ROI.

Sources

These are the project's **own illustrative early-stage estimates**, consolidated from the "Return on Investment" section of `mitsue_implementation_plan.md` — not externally audited figures. The underlying load, PUE, and carbon assumptions and their references (HIGHRESO PUE target, J-Credit methodology, Forestry Agency sequestration data, edge-data-center power benchmarks) are documented in that plan's Phase 1 "Sources & assumptions" block.

The categories above will be validated against METI FIT/FIP tariff schedules, J-Credit pricing, and biomass/timber market data **during** the Phase 1 feasibility studies (M9, December 2026), which will replace these framework figures with vetted estimates.

Rob Oudendijk — YR-Design / Safecast Mitsue, Nara Prefecture, Japan June 2026